📚LIBRARY MANAGEMENT SYSTEM USING SQL

2. DATABASE CREATION

CREATE DATABASE LIBRARYMANAGEMENT;

The above command creates a database named **LIBRARYMANAGEMENT** which is used to store all tables and data.

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3. TABLES USED

1. Authors Table

Stores details of authors like name and country.

2. Books Table

Stores book details and links with authors.

3. Members Table

Stores library member information like name, email, and phone.

4. BorrowRecords Table

Stores borrowing details of books by members including borrow and return dates.

4. TABLE CREATION

Authors Table

```
CREATE TABLE Authors (
   AuthorID INT IDENTITY(1,1) PRIMARY KEY,
   AuthorName VARCHAR(100) NOT NULL,
   Country VARCHAR(50)
);
```

Books Table

```
CREATE TABLE Books (
   BookID INT IDENTITY(1,1) PRIMARY KEY,
   Title VARCHAR(200) NOT NULL,
   AuthorID INT,
   Quantity INT,
   FOREIGN KEY (AuthorID) REFERENCES Authors(AuthorID)
);
```

Members Table

```
CREATE TABLE Members (
   MemberID INT IDENTITY(1,1) PRIMARY KEY,
   MemberName VARCHAR(100) NOT NULL,
   Email VARCHAR(100),
   Phone VARCHAR(15)
);
```

BorrowRecords Table

```
CREATE TABLE BorrowRecords (
   BorrowID INT IDENTITY(1,1) PRIMARY KEY,
   MemberID INT NOT NULL,
   BookID INT NOT NULL,
   BorrowDate DATE NOT NULL,
   ReturnDate DATE,
   FOREIGN KEY (MemberID) REFERENCES Members(MemberID),
   FOREIGN KEY (BookID) REFERENCES Books(BookID)
);
```

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5. DATA INSERTION

Authors Data

Inserted 30 authors including:

J.K. Rowling, George Orwell, Paulo Coelho, Agatha Christie, Dan Brown, etc.

Books Data

Inserted books like:

Harry Potter, 1984, The Alchemist, Sherlock Holmes, Sapiens, etc.

Members Data

Inserted 30 members with names, emails, and phone numbers.

Borrow Records Data

Inserted borrowing details linking members and books with borrow and return dates.

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6. BASIC QUERIES

SELECT * FROM Authors;

SELECT * FROM Books;

SELECT * FROM Members;

SELECT * FROM BorrowRecords;

These queries are used to display all records from respective tables.

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7. TABLE LIST QUERY

```
SELECT TABLE_NAME
FROM INFORMATION_SCHEMA.TABLES
WHERE TABLE_TYPE='BASE TABLE';
```

This query is used to display all tables present in the database.

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8. RESULT

The system successfully stores and retrieves data related to authors, books, members, and borrowing records using SQL queries.

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9. CONCLUSION

This project helped in understanding database creation, table relationships, data insertion, and SQL query execution.
It improved practical knowledge of relational database management using SQL.